NOVA-Q: Verified LLM Agents From Natural Language and Open Data to QAOA on CUDA-Q

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MOTIVATION & CONTRIBUTIONS

Problem. Turning a request (*"place 3 EV chargers in Da'an District, Taipei, to cover the most residents"*) into a QUBO needs data cleaning, formulation and penalty tuning expertise.

Prior work assumes a correct intermediate model [1], learns from user-given test cases [6], checks the translation on standard problem types [7], or uses a hand-built model [2].

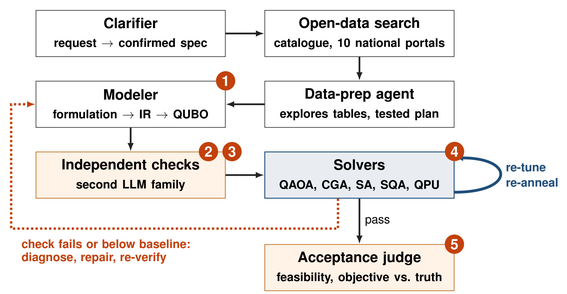
NOVA-Q (**N**atural-language → **O**pen-data → **V**erified **A**gents for **Q**uantum) contributes:

1 **Verify around every solve**: three gates before the solver, two after; the checker agent never sees the model.

2 **Cross-family checking**: a second LLM family catches 98% of injected data faults (same family: 73%).

3 **One verified QUBO, five solvers**, incl. QAOA on CUDA-Q.

WORKFLOW



1–5 gates. **Inner loop**: solver feedback re-tunes penalties and parameters. **Outer loop**: a failed check sends the model back for repair and re-verification; if repair fails, the verified baseline is delivered. Formulation agents from QuantumQUBO Agent [6]; the rest is new.

REAL CITIES, OFFICIAL OPEN DATA

One sentence in, e.g. *"Place 3 EV chargers in Da'an District to cover the most residents within 1 km."* 3 runs per case, independent judge.

City & task	official source	instance (real data)	passed
Taipei: EV chargers	village census	53 villages, 0.28 M people	3/3
Taipei: bike tour	YouBike live API	18 stations	2/3
Taipei: AED zones	AED registry	48 sites	3/3
London: ambulances	ONS Census 2021	27 areas, 0.21 M people	1/3
Berlin: playgrounds	Umweltatlas	36 sites, 0.29 M people	0/3
Passed, dev set, best of 2 LLM configs (other: 5/15)			**9/15**

METHODOLOGY

Benchmark set. Agent models that passed the judge, on their real data, shrunk with the pipeline's own point-reduction tool (weighted k-means) so the optimum is known by full enumeration.

Agent-built model	points	qubits
EV chargers (MCLP)	3–7	9–27
Bike tour (TSP)	3–5	9–25
Ambulance (p-median)	3–4	12–20
AED zoning (groups)	3, 6, 9	9–27

3 instances per size (k-means on random 80% subsets of the real points).

Map: 53 Da'an li (grey, size = residents), 7 representatives (○), optimal 3 chargers (■) with 1 km radius, solved exactly and by every solver.



QAOA on CUDA-Q. QUBO → Ising H, $|\gamma,\beta\rangle = \prod_{l=1}^{p} e^{-i\beta_l \sum_k X_k} e^{-i\gamma_l H} |+\rangle^{\otimes n}$, nvidia state-vector target, COBYLA, 3 restarts, 10^4 shots.

Other solvers. SA, SQA (OpenJij [3]; β tuned on 3 instances), D-Wave Advantage QPU: 1 000 reads; CGA: one 10 s job. Default penalty weights for all; one run each. Samples scored on the original problem.

VERIFICATION GATES

Our main contribution: gates 1–3 before solving, 4–5 after.

1 **Data schema & model check**: data contract; brute force on a small instance.

2 **Requirement check**: a blind checker agent's mutants must be rejected.

3 **Data cross-check** on the raw files by a second LLM family.

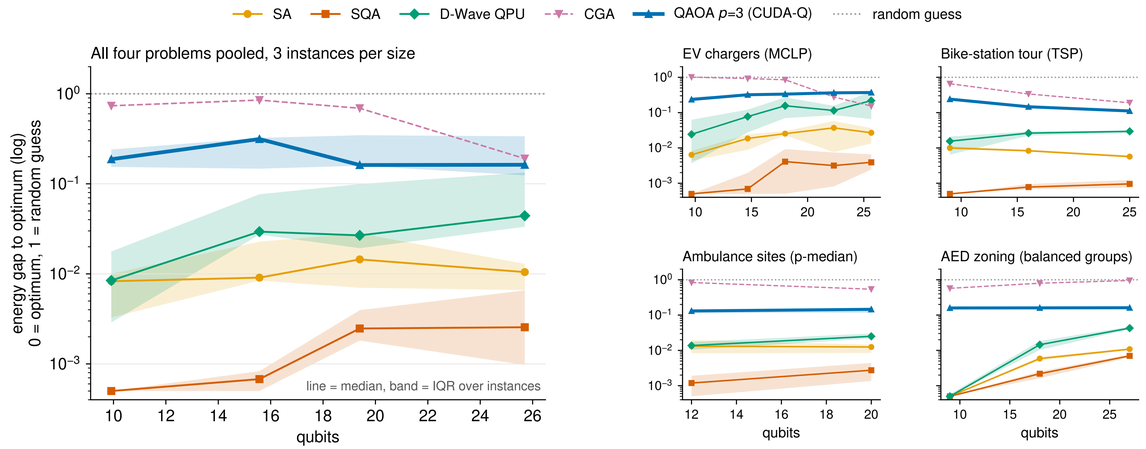
4 **Solution check** on the original constraints and a baseline.

5 **Independent judge** vs. hand-built ground truth.

	G1 data schema	G2 requirement	G3 cross-check	any gate
D1 entities missing	83%	6%	89%	**100%**
D2 duplicate place	50%	—	89%	**100%**
D3 swapped lat/lon	100%	—	56%	**100%**
D4 value off by 10x	0%	—	100%	**100%**
D5 total row mixed in	50%	—	89%	**100%**
D6 outside the area	50%	—	78%	**89%**
M1 constraint dropped	—	60%	20%	**60%**
M2 objective flipped	—	100%	17%	**100%**
M3 pair terms dropped	—	100%	—	**100%**
M4 count off by one	—	100%	17%	**100%**
M5 entity blocked	—	50%	17%	**67%**

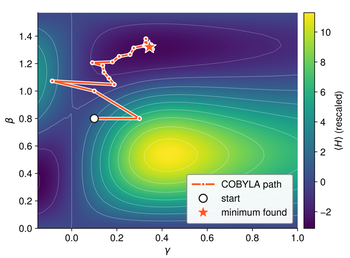
Injected harmful faults, n=9–18 per row; doer Qwen3.6-35B-A3B, checker Gemma. Hatched = outside the gate's scope. Caught: data 97/99, model 66/78. No false rejections in 15 verified runs (300/300 valid decisions).

RESULTS: ENERGY GAP TO THE OPTIMUM (LOWER IS BETTER)



Gap = $(E - E_{\text{opt}})/(\bar{E}_{\text{rand}} - E_{\text{opt}})$, 39 instances. Median: SQA 0.001, SA 0.009, QPU 0.028, QAOA 0.16, CGA 0.66. Optimum sampled: QAOA, SA, SQA, CGA 39/39; QPU 37/39.

QAOA ENERGY LANDSCAPE ON 2 GPUS



$\langle H \rangle(\gamma, \beta)$ at p=1 (for display; results use p=3), 26-qubit EV-charger instance.

1 600 CUDA-Q evaluations on 2× L40S (nvidia-mqpu) in **136 s**.

Orange: COBYLA path from the linear-ramp start into the low-energy valley.

CONCLUSION & FUTURE WORK

- Gates 1–3 stop wrong models before solving (98% / 85% of injected data / model faults).
- One verified QUBO, five solvers. QAOA on CUDA-Q samples the optimum on every instance, but trails the annealers: penalties cut its feasible fraction.
- Next: constraint-preserving mixers; nvidia-mgpu on NVLink GPUs (H200).

NVIDIA ACCELERATED STACK

- **LLM agents**: vLLM on 2× NVIDIA L40S (48 GB), FP8 weights: Gemma-4-31B-it (Google), Qwen3.6-35B-A3B and Qwen3.6-27B (Qwen); all Apache-2.0. No cloud API.
- **QAOA**: CUDA-Q 0.14, nvidia state-vector target (cuStateVec) on L40S.
- **GPU annealer**: Compal CGA [2].

QAOA cost evaluation (p=2, CUDA-Q 0.14)	qubits	seconds
CPU (qpp-cpu)	21	31.5
GPU, 1× L40S (nvidia)	21	**0.079** (**400**×)
GPU, 1× L40S, largest that fits	32	19.8

nvidia-mqpu, 2× L40S	qubits	1 GPU	2 GPUs
EV chargers (real data)	26	5.5	**9.0 eval/s**
Bike tour (real data)	25	4.3	**8.3 eval/s**

Parallel QAOA evaluations: 1.65–1.94× throughput. nvidia-mgpu over PCIe (no NVLink) was slower: NVLink systems (H200) are next.

REFERENCES

[1] H. Zhang, M. Emu, S. Choudhury, LLM-QUBO, AAAI-SS / arXiv:2509.00099 (2025). [2] Y. Chou *et al.*, GTC 2026 P81105. [3] OpenJij, github.com/OpenJij. [4] NVIDIA CUDA-Q 0.14. [5] E. Farhi *et al.*, QAOA, arXiv:1411.4028. [6] N. K. Mondal, M. R. Parvez, QuantumQUBO Agent, ICML 2026 Workshop, arXiv:2609.10629. [7] K. Sato *et al.*, IEEE QCE 2025.